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EWB Team Proposal

Assessment 3 – Design Forum Presentation

Group: Green Energy

Introduction

The Green Energy group has focused on the Design Area 3: Energy and the Project Opportunity, 3.2 Solar-powered coral terrarium. The aim of this proposal is to create a self-sustaining coral terrarium for close observation of coral for research, cultural and educational purposes.

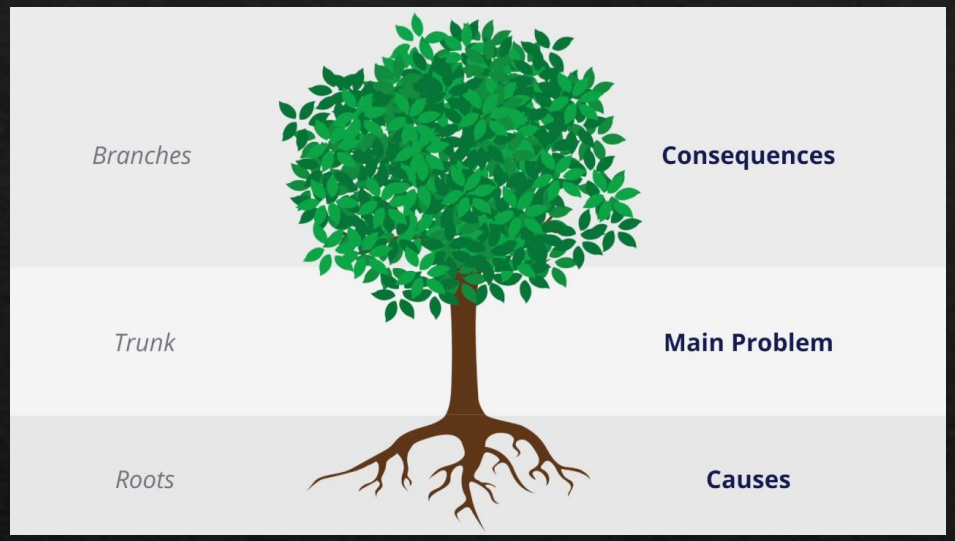


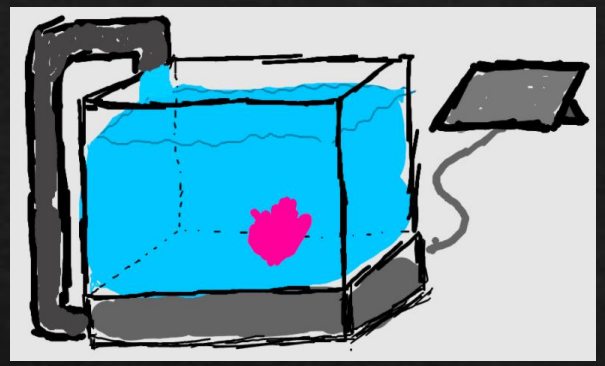
Figure 1: Problem Tree Analysis

How Might We...

With the influx of demand for research into climate change and its consequence of the ecological collapse of reefs around the world, there is a newfound necessity for coral specimens to be studied. Such a demand had yet to be facilitated by a portable self-sustaining terrarium design within the region our design area is targeting (Far North Queensland), leading to a shortage of corals specimens to observe and with limited access and portability of the current specimens in the area. Not only is there a large demand for corals for research purposes, but there is also a high demand for cultural and educational needs.

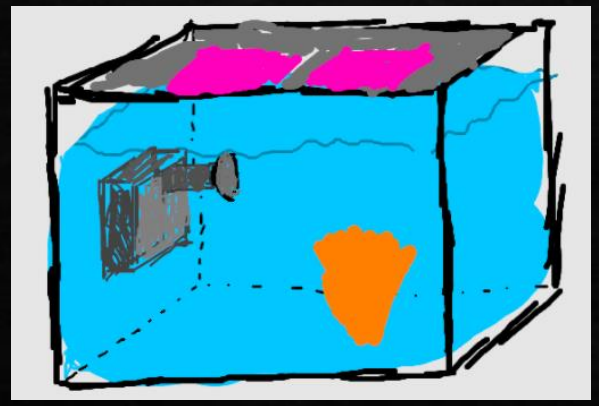
How might we preserve and display coral specimens for researchers and local Indigenous people in order to increase research, and make coral specimens more accessible and affordable to view and study?

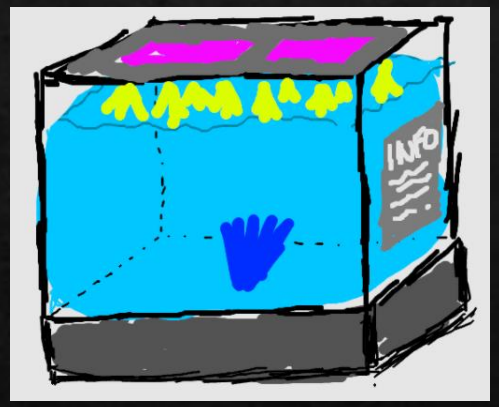
Design Solutions



Design 1:
Design one features an external solar panel to power the necessary components of the coral tank. The solar power will provide energy for the water pump to circulate water through the tank as well as power the life support necessary for coral to survive outside of their natural reef ecosystem.

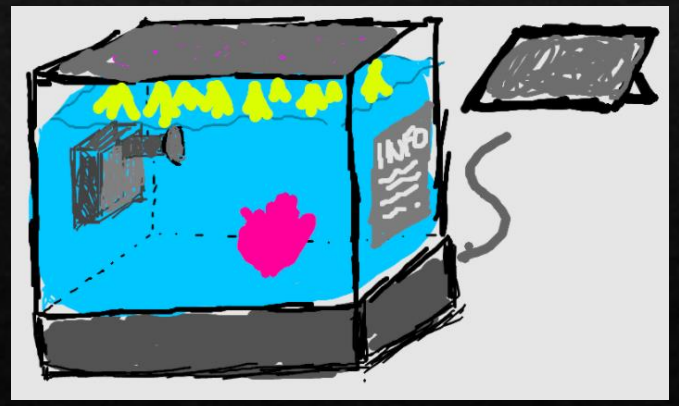
Design 2:
This design is one with the minimum necessary features requires to fulfill the design brief and needs of the community. Solar panels are connected to the top of the tank which power the small air pump as well as acts as a heater for the water.





Design 3:
Design three follows a more cultural appreciation design for the tank, including an information board detailing the current coral specimen(s) in the tank as well as the importance of coral and coral reef ecosystems to the Indigenous community. Solar panels are attached to the top of the tank, providing energy to the lighting system above and life support systems below the main section of the tank

Design 4:
Design four brought together the better components of the other three designs to form a design which was best served the purpose of the project, through a balanced solution. It contains all the necessary life support, solar panels to power it and an element of cultural appreciation.



Selected Design

Design 4: Synthesised Design

Through evaluating all designs and applying the scoring matrix method, our group concluded that the best design, which met the needs of the community and best fit the design brief, was Design 4.

Design 4, combined the best elements of other designs into one, creating a solution which was effective, energy efficient and provided cultural insights as well through an information board about the coral and effects of their ecosystems on the society and culture of the Indigenous people.

Similar to previous designs, this design holds the necessary life support in the base of the tank, providing heat, pH concentration, etc., for the coral. The solar panels on the design are external to allow the tank to be kept on display whilst the main energy provider can still collect solar power from outside.

Design 1 supplied the solution of an external solar panel so that the tank could be, for instance, kept inside for observation and research whilst the solar panel continued to function from outside, connecting the two via a cable.

Design 2 lacked too many features to be a suitable choice for the final design. As it didn't have much life support systems, and was essentially, just a plexiglass cube, Design 4 incorporated the simplicity of Design 2, adding on other elements such as from Design 1 and 3.

Unlike Design 3, this design has only the minimum necessary life sustaining components to cut down on costs and fit the requirements of just a temporary coral specimen storage tank. Design three, otherwise, had too many extra equipment which wasn't needed and increased the cost, complexity and power consumption.